AR Electric Field — Plane AR

Print the **stickers** below at 100% (each black pattern = 4 cm), lay them flat on a desk, open **Plane AR** in Safari and tap **Start camera**. The layout **auto-locks**: each point charge then **floats above** its sticker and a true 3-D electric field is drawn around it — no 3-D printer needed. Walk around to view; the field stays rigid (no jitter). Tap **Re-scan** for a new layout.

Stickers

Code	Marker	Role
0		Positive charge (+) — floats above its sticker
1		Negative charge (–)
6		Positive charge (+) — second one, for two like charges
7		Negative charge (–) — second one
2		Positive electron gun — fires e^+ along its printed arrow
3		Negative electron gun — fires e^-
4		Positive charged plate — vertical board, 4× edge, charge 2×
5		Negative charged plate — charge –2×

Ready-made experiments

Each of these is its own one-page printout (see the print menu): **opposite charges** (dipole), **like charges** (repel), **parallel plates** (capacitor), **a charge and a plate**, and **electron gun in a field**. The two schemes (Plane and Cube) reuse the same barcodes — run one at a time.

Tips: print at actual size, keep the white border (quiet zone) around each code, use even lighting on a matte surface. Tune with `?lift=` (charge height), `?plateside=`, `?plateq=`, `?gundir=`, `?smooth=`.